

Load-Independent Hybrid Resonant Converter for Automotive LED Driver Applications

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Abstract—In this article, a hybrid resonant LED driver for automotive illumination applications is proposed. The employed *LLC* resonant network provides load-independent output current characteristic and accomplishes full soft-switching operation of the semiconductor devices within the desired input voltage range. To provide constant frequency operation and nonpulsating input current, a continuous conduction mode boost structure is merged into the resonant network. Also, the analog dimming capability range of 50%–100% is attained for the LED string by adjusting the main switches operating duty cycle. Detailed theoretical analysis of the proposed LED driver is presented. To verify the analysis and effectiveness of the proposed topology, a 40-W laboratory prototype is designed and implemented for the input voltage range between 8–18 V.

Index Terms—LED driver, load-independent output current, resonant converter, soft-switching.

I. INTRODUCTION

HIGH brightness light-emitting diodes (HB-LEDs) are rapidly evolving to replace the traditional incandescent and fluorescent lamps in diverse lighting applications due to high luminous efficiency, compact size, and long lifespan [1]. In automotive applications, a compact size LED driver with high efficiency and low input current ripple should be employed for interfacing 12–14 V battery and HB-LED string. Over different operating conditions, battery voltage ranges from 8 to 18 V. Also, regarding the exponential current–voltage (*I*–*V*) characteristic of LEDs, optimum operating point and brightness are highly sensitive to the forward voltage. Therefore, the LED driver should support a well-regulated output current. Additionally, different automotive outside illuminations, such as headlights, brake lights, fog lights, and turn signals, demand different brightness levels, which require various LED string length. Hence, the employed LED driver should provide a wide output voltage range [2], [3].

Linear regulator-based LED drivers offer simple structure and low cost. In these topologies, constant current is maintained for LED string by utilizing a series transistor [4]–[6]. However, due

to high power dissipation, efficiency is low, and achieving a wide output voltage range requires a step-up pre-regulator [4]. Employing conventional buck-type and boost-type dc–dc converters are highly efficient approach for automotive LED drivers [7]–[9]. However, these topologies are inherently voltage-controlled circuits, and a tight current feedback is required to regulate the LED string current. In addition, the appropriate generated output voltage should be lower or higher than the input voltage to support a wide variety of LED string lengths, which is not possible with these topologies. A noninverting four-switch buck–boost LED driver is presented in [10]. To provide wide output voltage range, converter structure can be exchanged between the buck, boost, and buck–boost to provide a desired voltage gain with respect to the battery voltage level, but the converter efficiency drops in the buck–boost operation due to the cascaded structure. Cuk-based LED driver of Corradini and Spiazzi [11] employs coupled inductors and new digital control method, which provides the minimum phase dynamics and output current regulation without direct sensing the LED string current. However, the converter operation is hard switching. Cuk LED driver proposed in [12] can operate in a wide input and output voltage range with soft-switching performance. However, the converter design for tight regulation of the LED string current is challenging due to the nonminimum phase dynamic essence of the Cuk converter.

In recent research work, resonant converters have attracted more interest due to their inherent soft-switching capability, high efficiency, and high power density [13]–[15]. An *LLC*-based resonant LED driver for automotive applications is presented in [16]. To provide high output voltage for driving 30 series LEDs in a string, a boost converter is utilized as a front-end stage. However, two individual transformers are employed and a frequency control scheme is required for regulating the LED string current. To achieve constant frequency control, a buck–boost converter is merged into a reconfigurable series resonant converter [17]. Main inverter switches of this topology can be altered between full-bridge and half-bridge structures depending on the battery voltage level. However, due to the cascaded structure, the efficiency is reduced. Particular resonant structures, such as *LCL*, *LC3L*, and *LCLC*, called immittance networks, can provide load-independent output current, which is well suited for LED driver applications [18]–[21]. Since the load-independent current operation is attained at the resonant frequency, constant frequency methods, such as phase-shift (PS) control, are employed for *LCL*-based resonant converters of the works in [19]–[21] to regulate the LEDs current at the desired

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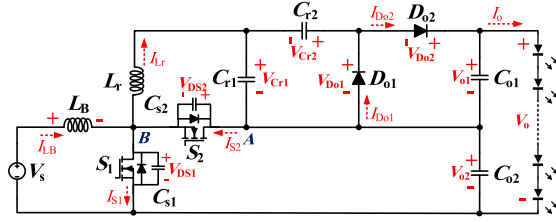


Fig. 1. Circuit diagram of the proposed LED driver.

level; but these schemes require at least four switches, and the circulating current is increased due to the PS control.

In this article, an *LCC*-based resonant converter with inherent load-independent output current characteristic is proposed for automotive LED driver applications. In order to provide low input current ripple, well-regulated output current within the automotive battery voltage range, and employ the conventional pulsewidth modulation (PWM) control, a continuous conduction mode (CCM) boost converter is combined with the modified *LCC* resonant network. Due to the semicascaded structure of the proposed topology, the converter efficiency is improved in comparison to the conventional cascaded counterparts [22]. Additionally, the resonant network can provide zero-voltage-switching (ZVS) operation for the main switches and alleviates the reverse recovery of the rectifying output diodes within the desired input voltage range. Furthermore, the dimming range of 50%–100% is arranged by adjusting the main switches duty cycle.

The rest of this article is organized as follows. The proposed LED driver operation principles and the steady-state analysis are presented in Sections II and III, respectively. The experimental results of a 40-W prototype are provided in Section IV. Finally, Section V concludes this article.

II. PROPOSED LED DRIVER AND OPERATING PRINCIPLES

A. Topology Description

In Fig. 1, the schematic diagram of the proposed hybrid resonant LED driver is illustrated. This converter is comprised of a modified *LCC* resonant network, rectifying output diodes, and a front-end boost converter. Nonisolated resonant *LCC* converter consists of main switches (S_1 and S_2), resonant network (L_r , C_{r1} , and C_{r2}), two rectifying output diodes (D_{o1} and D_{o2}), and output capacitor C_{o1} . The CCM boost converter components are L_B , S_1 , S_2 body diode, and output capacitor C_{o2} . At the load side, C_{o1} and C_{o2} are stacked, which can provide high output voltage for driving a series-connected LED string. Also, due to the stacked output structure, a portion of the C_{o2} energy is directly delivered to the load, which can improve the converter efficiency. The switching frequency (f_s) is close to the resonant frequency of the *LCC* resonant network (f_0) and the output current is regulated by the S_1 duty cycle.

B. Operation Principles

To facilitate the converter operation, all components are assumed to be ideal and the output capacitors are assumed to be

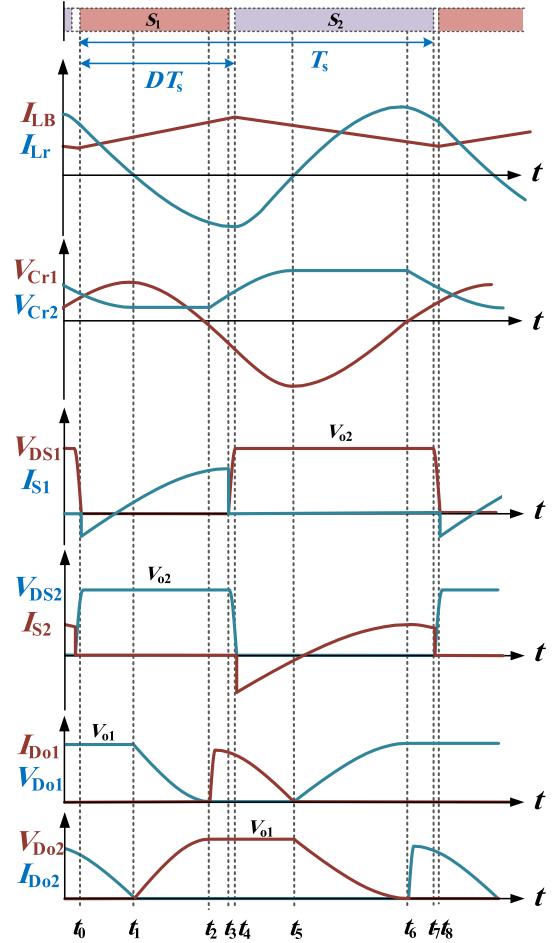


Fig. 2. Key waveforms of the proposed converter.

large enough so that their voltages can be considered constant during a switching cycle. Also, for accurate analysis of the converter operation, the main switches output capacitances (C_{s1} and C_{s2}) are considered. The proposed converter operation is divided into eight intervals. The key waveforms are presented in Fig. 2, and the converter equivalent circuit in each interval is shown in Fig. 3.

Interval 1 [t_0 – t_1]: This interval begins by turning S_1 ON while D_{o1} is reversed biased and D_{o2} is conducting. L_B is charged via V_s , whereas V_{o2} is applied to the resonant tank input side (V_{AB}). I_{Lr} and V_{Cr2} decrease, but V_{Cr1} increases. At t_1 , I_{Lr} reaches zero and D_{o2} turns OFF at zero current switching (ZCS).

Interval 2 [t_1 – t_2]: During this interval, resonance continues between L_r and C_{r1} , therefore, C_{r1} and L_r discharge in a sinusoidal fashion. At t_2 , ($V_{Cr1} + V_{Cr2}$) amplitude reaches zero and D_{o1} turns ON.

Interval 3 [t_2 – t_3]: In this interval, S_1 and D_{o1} are conducting. A resonance occurs between L_r , C_{r1} , and C_{r2} , hence, C_{r1} and C_{r2} discharge and charge, respectively. At t_3 , S_1 is turned OFF and this mode ends.

Interval 4 [t_3 – t_4]: By turning S_1 OFF, the summation of I_{LB} and I_{Lr} charges and discharges C_{s1} and C_{s2} , respectively. At t_4 , C_{s2} is completely discharged and

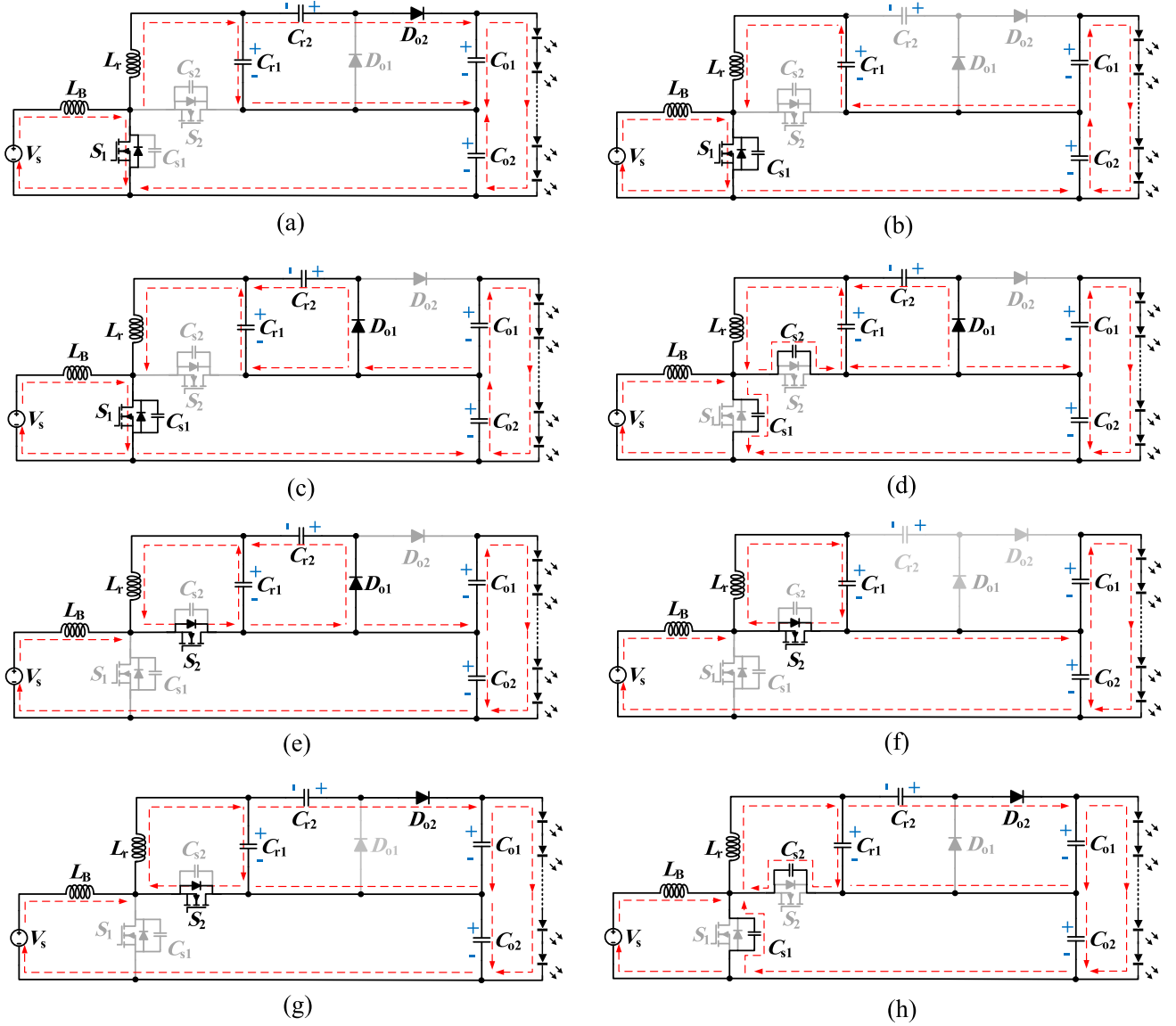


Fig. 3. Equivalent circuit of each operating interval. (a) Interval 1. (b) Interval 2. (c) Interval 3. (d) Interval 4. (e) Interval 5. (f) Interval 6. (g) Interval 7. (h) Interval 8.

D_{s2} turns ON. Hence, S_2 can be turned ON under ZVS conditions.

Interval 5 $[t_4-t_5]$: During this interval, S_2 and D_{o1} are conducting, (V_s-V_{o2}) is placed across L_B , and I_{LB} decreases linearly. Also, zero voltage is applied to the resonant tank input. At t_5 , I_{Lr} reaches zero and D_{o1} turns OFF at ZCS.

Interval 6 $[t_5-t_6]$: At t_5 , S_2 is ON and rectifying output diodes are in the OFF state. L_B is demagnetizing and the resonance continues between L_r and C_{r1} , which charges C_{r1} through I_{Lr} . At t_6 , $(V_{Cr1}+V_{Cr2})$ reaches V_{o1} , and D_{o2} turns ON.

Interval 7 $[t_6-t_7]$: During this interval, S_2 and D_{o2} are conducting, C_{r2} participates in the resonance and discharges through D_{o2} . At t_7 , S_2 is turned OFF and this mode ends.

Interval 8 $[t_7-t_8]$: By turning S_2 OFF, the difference of currents I_{LB} and I_{Lr} discharges and charges C_{s1} and C_{s2} , respectively. At t_8 , C_{s1} is completely discharged and D_{s1} turns ON. Hence, S_1 can be turned ON under ZVS condition.

III. STEADY-STATE ANALYSIS AND DESIGN CONSIDERATION

A. Conversion Ratio

The steady-state analysis of the proposed converter is carried out with the assumption of ideal semiconductor devices. Since a CCM boost converter is employed at the front-end stage, the voltage across C_{o2} is obtained by applying the volt-second principle on the input inductor L_B as follows:

$$V_{o2} = \frac{V_s}{1-D} \quad (1)$$

where D is the operating duty cycle of S_1 . The voltage across the switch S_2 is applied to the LCC resonant network (V_{AB}), which is an asymmetric square waveform with V_{o2} amplitude. Since the proposed converter operates at the adjacent of the resonant frequency, the fundamental harmonic approximation

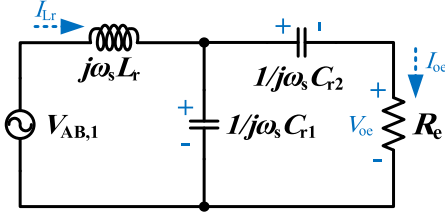


Fig. 4. FHA equivalent circuit of the resonant network.

(FHA) approach can be adopted with proper accuracy for theoretical analysis [23], [24]. The fundamental frequency amplitude of V_{AB} is derived as

$$V_{AB}^1 = \frac{V_s}{\pi(1-D)} \sqrt{2(1 - \cos(2\pi D))} \sin(\omega_s t + \phi) \quad (2)$$

where

$$\omega_0 = \frac{1}{\sqrt{L_r C_{r1}}}, \phi = \tan^{-1} \left(\frac{\sin(2\pi D)}{1 - \cos(2\pi D)} \right). \quad (3)$$

In Fig. 4, the *LCC* resonant network ac equivalent circuit at the fundamental frequency is shown. Based on this figure, the voltage gain $M(j\omega_n)$ and the current gain $H(j\omega_n)$ are obtained as

$$M(j\omega_n) = \frac{V_{oe}}{V_{AB}^1} = \frac{jC_n \omega_n}{Q_e (1 - \omega_n^2 (1 + C_n)) + jC_n \omega_n (1 - \omega_n^2)} \quad (4)$$

$$H(j\omega_n) = \frac{I_{oe}}{\left(\frac{V_{AB}^1}{Z_0}\right)} = \frac{jC_n \omega_n}{1 - \omega_n^2 (1 + C_n) + jC_n Q_e \omega_n (1 - \omega_n^2)} \quad (5)$$

where the normalized frequency ω_n , the resonant capacitors ratio C_n , the characteristic impedance Z_0 , and the quality factor Q_e are defined as

$$\omega_n = \frac{\omega_s}{\omega_0} = \frac{f_s}{f_0}, C_n = \frac{C_{r2}}{C_{r1}} \quad (6)$$

$$Z_0 = \sqrt{\frac{L_r}{C_{r1}}}, Q_e = \frac{Z_0}{R_e} = \frac{\omega_0 L_r}{R_e}. \quad (7)$$

Since the rectifying stage of the resonant network is derived from the conventional voltage doubler [24], the equivalents output resistance of the *LCC* network R_e is obtained as

$$R_e = \frac{2}{\pi^2} \frac{NV_D}{I_o} \left(1 - \frac{V_s}{NV_D(1-D)} \right) \quad (8)$$

where N is the number of LEDs in a string, and V_D is the LED threshold voltage. The magnitude plot of (5) at different Q_e and C_n values is illustrated in Fig. 5(a) and (b), respectively. As can be observed, the behavior of *LCC* network is almost the duality of the conventional *LLC* network. It can be found that at operating points in the vicinity of $\omega_n = 1$, a load-independent output current characteristic is achieved. Also, at this point, the tolerance of C_n has no impact on the output current. Using (2) and (5), the output current of the proposed converter is obtained

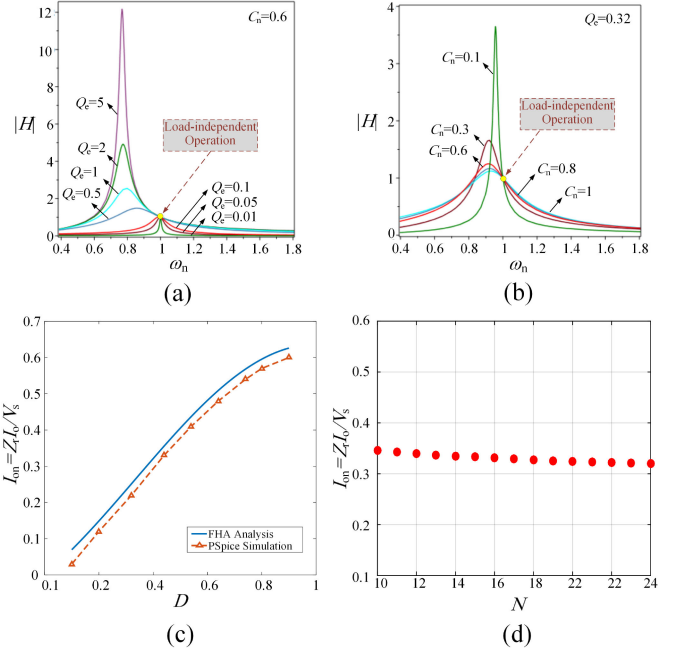


Fig. 5. Theoretical current gain graphs. (a) *LCC* resonant network at different quality factors. (b) *LCC* resonant network at different resonant capacitors ratios. (c) Normalized output current as a function of duty cycle. (d) Normalized output current as a function of the number of LEDs in a string.

as follows:

$$I_o = \frac{1}{\pi^2} \frac{V_s}{Z_0} \frac{\sqrt{2(1 - \cos(2\pi D))}}{1 - D}. \quad (9)$$

According to (9), in Fig. 5(c), the normalized current of the LED string versus the S_1 operating duty cycle is plotted and compared with the simulation results. Since the proposed converter operates near the resonant frequency of the *LCC* network, the employed FHA method has an acceptable accuracy for theoretical analysis. In Fig. 5(d), the output current variation versus the number of LEDs in a string is illustrated. As can be observed, the proposed converter has an excellent output current-source characteristic. It should be noted that, according to the operating modes, the minimum output voltage of the proposed converter is limited, which can be expressed as

$$V_o > \frac{V_{s(\min)}}{1 - D} + 2V_{Do} \quad (10)$$

where V_{Do} is the forward voltage drop of D_{o1} and D_{o2} .

B. Semiconductors Voltage and Current Stresses

According to the operating intervals, the voltage stress of the main switches is limited to V_{o2} , also, the voltage stress across D_{o1} and D_{o2} is equal to

$$V_{Do1} = V_{Do2} = V_{o1} = NV_D - \frac{V_s}{1 - D}. \quad (11)$$

Employing FHA analysis, I_{LB} can be approximated with sinusoidal waveform with peak amplitude of

$$I_{LB_max} = \frac{P_o}{\eta V_s} + \frac{4DV_s}{\pi^2 L_B f_s} \quad (12)$$

where P_o is output power and η is the converter efficiency. The input current of the resonant network I_{Lr} can be expressed as

$$I_{Lr}(t) = \frac{V_{AB}^1}{Z_0} \sqrt{\frac{1}{Q_e^2} + \frac{(1 + C_n)^2}{C_n^2}} \sin(\omega_0 t + \phi + \alpha) \quad (13)$$

where

$$\alpha = \tan^{-1} \left(Q_e \left(\frac{1 + C_n}{C_n} \right) \right). \quad (14)$$

Since both I_{LB} and I_{Lr} flow in S_1 and S_2 during DT_s and $(1-D)T_s$ intervals, respectively, according to (12) and (13), the values of L_B and C_n affect the main switches current stress. Small values of C_n results in a large amplitude of I_{Lr} , which increases the main switches current stress. Also, small values of L_B lead to higher current stress of S_1 and S_2 . The average and peak current values of the output diodes are obtained as

$$I_{Do1_max} = I_{Do2_max} = \pi I_o \quad (15)$$

$$I_{Do1_DC} = I_{Do2_DC} = I_o. \quad (16)$$

C. Soft-Switching Conditions

Since a CCM boost converter is combined with the resonant network, both input inductor current I_{LB} and the resonant inductor current I_{Lr} flow through the main switches. According to the operating intervals, at $t = t_3$, I_{Lr} current is negative, and $(I_{Lr} - I_{LB})$ current can provide a strong ZVS turn-ON condition for S_2 due to the enough stored energy in both L_B and L_r . However, at $t = t_7$, I_{Lr} current is positive and the ZVS turn-ON condition for S_1 is achieved by $(I_{LB} - I_{Lr})$ current. As a result, the ZVS turn-ON criteria is more critical for S_1 , which can be expressed as follows:

$$C_s \left(\frac{V_s}{1-D} \right)^2 < \frac{1}{2} L_r (I_{Lr}(t_7) - I_{LB}(t_7))^2 \quad (17)$$

where $C_{s1} = C_{s2} = C_s$ and

$$I_{LB}(t_7) = \frac{P_o}{\eta V_s} - \frac{DV_s}{2L_B f_s}. \quad (18)$$

To ensure the ZVS turn-ON condition for S_1 , (17) must be satisfied for the entire input voltage and operating duty cycles. Just as the similar resonant converters, ZVS operation is realized at frequencies above resonance. But, to attain both load-independent output current feature and proper ZVS operation, the switching frequency is selected slightly above the resonant frequency. According to (17), ZVS region of S_1 is illustrated in Fig. 6. As can be observed, small values of C_n provide a better soft-switching range. However, based on (13), the conduction losses of the main switches would increase. As a result, a tradeoff should be performed between the soft-switching range and the conduction power losses.

IV. EXPERIMENTAL VERIFICATION

In order to evaluate the feasibility of the proposed resonant LED driver, a 40-W laboratory prototype with 8–18 V input voltage range, operating at 200-kHz switching frequency is designed and implemented. 24 series HB-LEDs from OSRAM

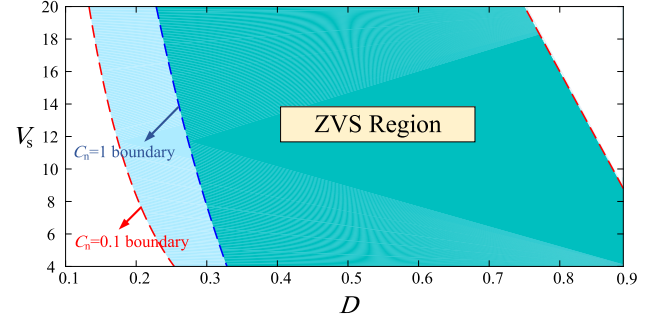


Fig. 6. Soft-switching boundaries of S_1 in terms of the input voltage versus the operating duty cycle and the resonant capacitors ratio.

with specifications 3.3 V/500 mA are used for the LED string. As a result, the nominal output voltage of the proposed converter is about 80 V.

According to the aforementioned considerations, the values of L_B , L_r , C_{r1} , and C_{r2} are designed. Based on Fig. 5(a), to enjoy the load-independent output current characteristic of the LCC resonant network, the operating point should be selected close to $\omega_n = 1$. Using (9) and considering the input voltage range of 8–18 V, parameter $Z_0 = 7.7 \Omega$ is obtained. As a result, the values of L_r and C_{r1} are calculated as

$$L_r = \frac{Z_0}{2\pi f_s} = 6.1 \mu\text{H} \quad (19)$$

$$C_{r1} = \frac{1}{2\pi f_s Z_0} = 104 \text{ nF}. \quad (20)$$

According to Fig. 6, C_n value determines the ZVS range of the main switches. In order to achieve ZVS turn-ON condition of S_1 within the entire operating range with minimized circulating current, the value of C_n is chosen about 0.6. As a result, $C_{r2} = 62 \text{ nF}$ is calculated. The value of L_B is designed just as the conventional boost converter for current ripple of 0.4 A as follows:

$$L_B = \frac{V_s D}{f_s \Delta I_{LB}} = 80 \mu\text{H}. \quad (21)$$

The output capacitors C_{o1} and C_{o2} are selected 30 μF for achieving about 30-mA output current ripple. However, to provide almost dc output current with smaller output capacitors, an extra inductor can be used in series with the LED string [21]. The complete specifications of the prototype converter are listed in Table I. A comparison between the proposed converter and recently reported resonant-based automotive LED drivers presented in [16], [17], [19], [21], [25], and [26] is summarized in Table II. It should be noted that an input capacitor is considered for topologies having pulsating input current. As stated in this table, the proposed converter has a lower components count than the other counterparts. The nonpulsating, continuous input current feature of the proposed converter is well suited for battery operation. Also, due to the constant frequency operation and load-independent output current feature of the proposed topology, the output current can be regulated by using the main switches operating duty cycle. In comparison with cascaded topologies of Wang *et al.* [16] and Veeramallu *et al.* [17], the

TABLE I
IMPLEMENTED PROTOTYPE PARAMETERS

Parameter/Component	Symbol	Specification
DC Input voltage	V_s	8-18 V
Output power	P_o	40 W
Output Current	I_o	0.5 A
Switching Frequency	f_s	200 kHz
Boost inductor	L_B	80 μ H
Resonant inductor	L_r	6.1 μ H
First Resonant capacitor	C_{r1}	104 nF
Second Resonant capacitor	C_{r2}	62 nF
Output capacitors	C_{o1}, C_{o2}	30 μ F
Main switches	S_1, S_2	IRFB3507
Output diodes	D_{o1}, D_{o2}	SB580

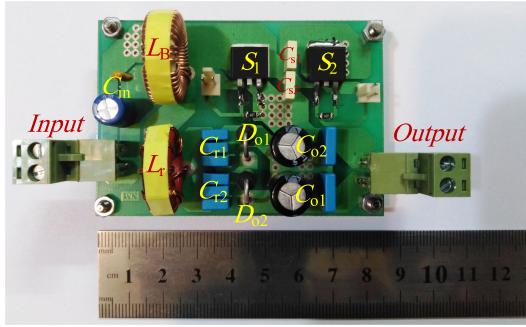


Fig. 7. Prototype setup photograph.

proposed converter efficiency is improved due to the semicascaded structure. The PS modulation employed in [19] and [21] increases the component count and complexity while imposing an extra circulating current, which results in higher losses when the converter operates over a wide input voltage range. The resonant LED drivers presented in [25] and [26] can provide load-independent output current, which is suitable for automotive lighting applications. However, these converters regulate the output current using variable frequency control, which cannot be employed in wide input voltage range applications due to the wide frequency range variations.

A photograph of the experimental prototype setup is illustrated in Fig. 7. The experimental operating waveforms of the semiconductor devices for 8- and 18-V input voltages are presented in Figs. 8 and 9, respectively. According to Figs. 8(a), (b), 9(a), and (b), the ZVS turn-ON condition of the main switches is maintained for the entire input voltage range. Also, as can be observed from Figs. 8(c), (d), 9(c), and (d), ZCS operation is realized for the output diodes, which alleviates the reverse recovery loss. The current waveforms of L_B and L_r for high and low input voltages are shown in Fig. 10. As can be observed, L_B functions in CCM, and the maximum input current ripple is below 400 mA, which is preferable for battery operation. Fig. 11 illustrates the output current and voltage waveforms during analog dimming conditions of 50%–100% brightness and

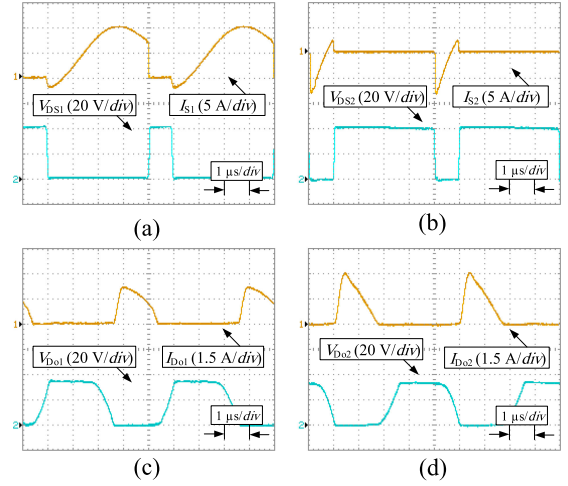


Fig. 8. Experimental waveforms of main switches and output diodes at 8-V input voltage. (a) S_1 . (b) S_2 . (c) D_{o1} . (d) D_{o2} .

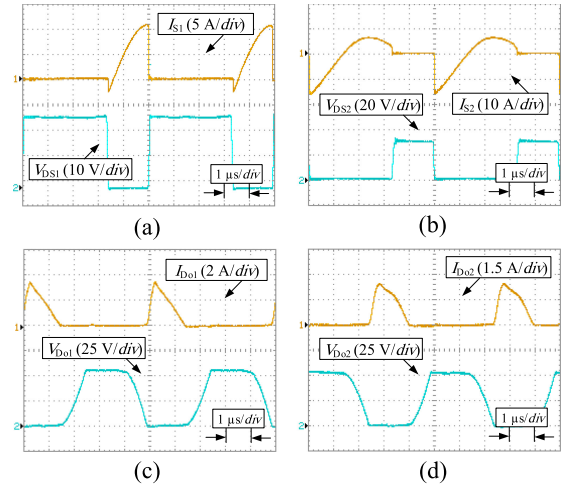


Fig. 9. Experimental waveforms of main switches and output diodes at 18-V input voltage. (a) S_1 . (b) S_2 . (c) D_{o1} . (d) D_{o2} .

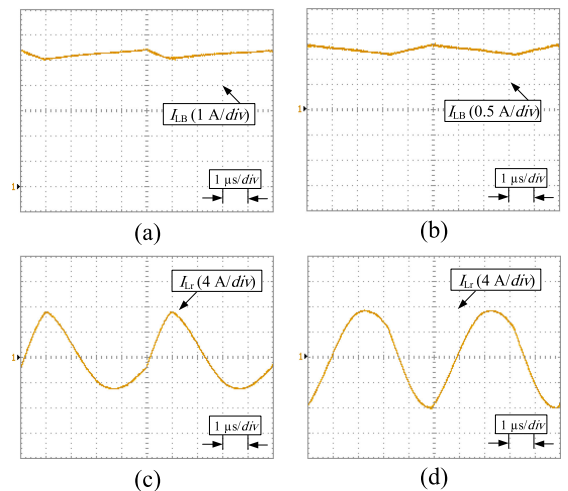


Fig. 10. Experimental waveforms of (a) I_{LB} at 8 V, (b) I_{LB} at 18 V, (c) I_{Lr} at 8 V, and (d) I_{Lr} at 18 V.

TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART LED DRIVERS

Characteristics		Proposed	[16]	[17]	[19]	[21]	[25]	[26]
Number of	Switches	2	2	4	4	6	2	2
	Diodes	2	2	4	-	-	4	6
	Inductors	2	5	2	2	3	1	3
	Capacitors	4	6	3	4	4	3	2
Voltage stress	Main switches	$\frac{V_s}{1-D}$	$\frac{V_s}{1-D}$	$\frac{V_s}{1-D}$	V_s	V_s	V_s	$\frac{2V_s}{M}$
	Output diodes/switches	$V_o - \frac{V_s}{1-D}$	V_o	$\frac{V_o}{2}$	V_o	V_o	V_o	$\frac{n_s V_s}{n_p M}$
Soft-switching		ZVS	ZVS	ZVS	ZVS	ZVS	ZCS	Almost ZVS
Operation		PWM	VFC**	PWM	PSC*	PSC*	VFC**	VFC*
Continuous input current		Yes	Yes	No	No	No	No	Yes
Reported full load efficiency (%)		96	91	92	91	85	93	94
Nominal power (W)		40	30	22	25	20	36	60

* Phase Shifted Control.

** Variable Frequency Control.

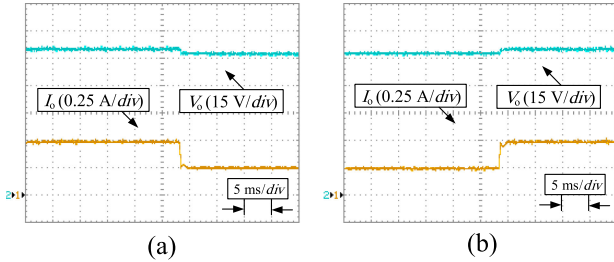


Fig. 11. LED string voltage and current waveforms during analog dimming: (a) 100% to 50% and (b) 50% to 100%.

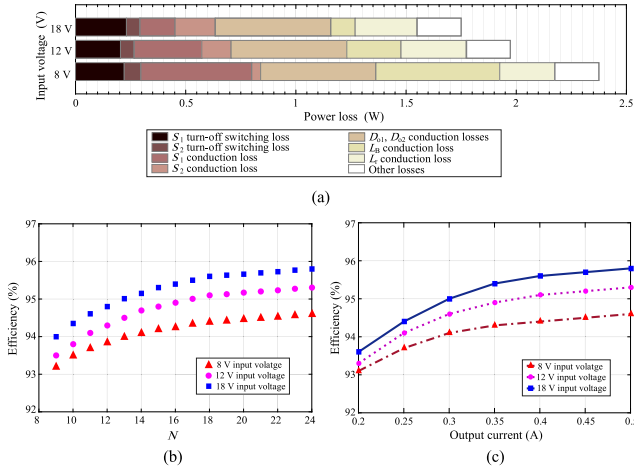


Fig. 12. (a) Loss distribution analysis at three different input voltages. (b) Efficiency plot for different number of LEDs. (c) Efficiency plot for different output currents.

vice versa. It should be noted that to provide a wide dimming range, the conventional ON–OFF control using a low-frequency PWM dimming signal can be employed just as the similar counterparts [17]. In Fig. 12(a), the detailed loss analysis for three different input voltages is presented. It should be noted that to mitigate the switching turn-OFF loss of S_1 and S_2 , a 1-nF snubber capacitor is employed. The efficiency diagrams of the proposed converter as a function of the output voltage and output current are presented in Fig. 12(b) and (c), respectively.

As can be seen, the overall efficiency of the proposed converter is about 95% for the entire input voltage range, and the highest efficiency of 95.8% is obtained at 18-V input voltage and full load condition.

V. CONCLUSION

The proposed hybrid resonant LED driver is well suited for automotive lighting applications. This converter enjoys load-independent output current feature and constant frequency operation. The combined CCM boost structure with the *LCC* resonant network provides low input current ripple and analog dimming capability range of 50%–100% using the main switches duty cycle. Full soft-switching operation of the main switches and output diodes are achieved within the desired input voltage range. The proposed topology is analyzed and designed based on the FHA method and the performance is verified by the implementation of a 40-W prototype operating at 200 kHz delivering 500 mA to a string of 24 HB-LEDs. Using regular semiconductor devices, the proposed converter achieves an overall efficiency of 93%–96% within the input voltage range of 8 to 18 V and over the wide output voltage range of 30 to 80 V. By a detailed loss breakdown analysis, the loss distribution of the proposed topology at different input voltages is demonstrated.

REFERENCES

- [1] Y. Wang, J. M. Alonso, and X. Ruan, "A review of LED drivers and related technologies," *IEEE Trans. Ind. Electron.*, vol. 64, no. 7, pp. 5754–5765, Jul. 2017.
- [2] M. Alexander, "Automotive LED lighting needs special drivers," *Power Electron. Technol.*, pp. 22–28, 2005.
- [3] S. Mukherjee, V. Yousefzadeh, A. Sepahvand, M. Doshi, and D. Maksimovic, "A two-stage automotive LED driver with multiple outputs," *IEEE Trans. Power Electron.*, vol. 36, no. 12, pp. 14175–14186, Dec. 2021.
- [4] Y. Hu and M. M. Jovanovic, "LED driver with self-adaptive drive voltage," *IEEE Trans. Power Electron.*, vol. 23, no. 6, pp. 3116–3125, Nov. 2008.
- [5] H.-J. Chiu, Y.-K. Lo, J.-T. Chen, S.-J. Cheng, C.-Y. Lin, and S.-C. Mou, "A high-efficiency dimmable LED driver for low-power lighting applications," *IEEE Trans. Ind. Electron.*, vol. 57, no. 2, pp. 735–743, Feb. 2010.
- [6] X. Lyu, Y. Li, G. Liu, and D. Cao, "A high-efficiency linear LED driver with concave current control for low-power application," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 3, no. 3, pp. 691–697, Sep. 2015.

- [7] A. Sureshkumar and R. Gunabalan, "Design and implementation of single switch control DC-DC converter with wide input variation in automotive LED lighting," *Int. Trans. Elect. Energy Syst.*, vol. 31, no. 4, 2021, Art. no. e12776.
- [8] A. Pollock, H. Pollock, and C. Pollock, "High efficiency LED power supply," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 3, no. 3, pp. 617–623, Sep. 2015.
- [9] J.-M. Wang, S.-T. Wu, W.-Y. Su, and Y.-L. Lin, "Study and implementation of the LED headlight driver with auto-start function in specific location," *IET Intell. Transport Syst.*, vol. 10, no. 10, pp. 623–634, 2016.
- [10] Y. Qin, S. Li, and S. Hui, "Topology-transition control for wide-input-voltage-range efficiency improvement and fast current regulation in automotive LED applications," *IEEE Trans. Ind. Electron.*, vol. 64, no. 7, pp. 5883–5893, Jul. 2017.
- [11] L. Corradini and G. Spiazzi, "A high-frequency digitally controlled LED driver for automotive applications with fast dimming capabilities," *IEEE Trans. Power Electron.*, vol. 29, no. 12, pp. 6648–6659, Dec. 2014.
- [12] A. Sepahvand, M. Doshi, V. Yousefzadeh, J. Patterson, K. K. Afridi, and D. Maksimović, "High-frequency ZVS Čuk converter for automotive LED driver applications using planar integrated magnetics," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, 2017, pp. 2467–2474.
- [13] M. F. Menke, J. P. Duranti, L. Roggia, F. E. Bisogno, R. V. Tambara, and Á. R. Seidel, "Analysis and design of the LLC LED driver based on state-space representation direct time-domain solution," *IEEE Trans. Power Electron.*, vol. 35, no. 12, pp. 12686–12701, Dec. 2020.
- [14] H. Dong, X. Xie, and D. Yu, "Primary-side regulation scheme for LLC resonant converter with improved resonant current sampling circuit," *IEEE Trans. Power Electron.*, vol. 36, no. 2, pp. 1242–1246, Feb. 2021.
- [15] F. Pouladi, H. Farzanehfard, E. Adib, and H. L. Sage, "Single-switch soft-switching LED driver suitable for battery-operated systems," *IEEE Trans. Ind. Electron.*, vol. 66, no. 4, pp. 2726–2734, Apr. 2019.
- [16] Y. Wang, S. Gao, Y. Guan, J. Huang, D. Xu, and W. Wang, "A single-stage LED driver based on double LLC resonant tanks for automobile headlight with digital control," *IEEE Trans. Transport. Electrification*, vol. 2, no. 3, pp. 357–368, Sep. 2016.
- [17] V. K. S. Veeramallu, S. Porpandiselvi, and B. Narasimharaju, "A non-isolated wide input series resonant converter for automotive LED lighting system," *IEEE Trans. Power Electron.*, vol. 36, no. 5, pp. 5686–5699, May 2020.
- [18] M. Borage, K. Nagesh, M. Bhatia, and S. Tiwari, "Resonant imittance converter topologies," *IEEE Trans. Ind. Electron.*, vol. 58, no. 3, pp. 971–978, Mar. 2011.
- [19] M. Khatua *et al.*, "High-performance megahertz-frequency resonant DC-DC converter for automotive LED driver applications," *IEEE Trans. Power Electron.*, vol. 35, no. 10, pp. 10396–10412, Oct. 2020.
- [20] S. Mukherjee, A. Sepahvand, and D. Maksimović, "High-frequency LC³ L resonant DC-DC converter for automotive LED driver applications," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, 2018, pp. 797–802.
- [21] S. Mukherjee, V. Yousefzadeh, A. Sepahvand, M. Doshi, and D. Maksimović, "High frequency wide range resonant converter operating as an automotive LED driver," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 5, pp. 5781–5794, Oct. 2020.
- [22] C. Li and J. A. Cobos, "Classification of differential power processing architectures based on VA area modeling," *IEEE Trans. Emerg. Sel. Topics Power Electron.*, to be published, doi: [10.1109/JESTPE.2021.3093654](https://doi.org/10.1109/JESTPE.2021.3093654).
- [23] D. J. Tschirhart and P. K. Jain, "A CLL resonant asymmetrical pulsedwidth-modulated converter with improved efficiency," *IEEE Trans. Ind. Electron.*, vol. 55, no. 1, pp. 114–122, Jan. 2008.
- [24] M. K. Kazimierczuk and D. Czarkowski, *Resonant Power Converters*. Hoboken, NJ, USA: Wiley, 2012.
- [25] L. Yang, W. Yu, and J. Zhang, "Variable frequency constant current control method for switched-capacitor converter based automotive LED driver," *IEEE Access*, vol. 7, pp. 42094–42106, 2019.
- [26] M. M. Ghahderijani, M. Castilla, A. Momeneh, J. T. Miret, and L. G. de Vicuna, "Frequency-modulation control of a DC/DC current-source parallel-resonant converter," *IEEE Trans. Ind. Electron.*, vol. 64, no. 7, pp. 5392–5402, Jul. 2017.



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